



# **FUTURE OF RELIABLE NETWORKING BEYOND THE TRADITIONAL OSI MODEL**

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# INTRODUCTION

- The OSI Model has been the foundation of computer networking for decades.
- Modern technologies like **5G, IoT, Cloud Computing, AI, and SDN** require more intelligent, flexible, and reliable networking approaches.
- Future networking moves beyond fixed layers to adaptive and software-driven architectures.

# **Traditional OSI Model and Its Limitations**

- **OSI Model Layers**

- Physical
- Data Link
- Network
- Transport
- Session
- Presentation
- Application

# **LIMITATIONS**

- Rigid layered architecture
- High latency due to layer-by-layer processing
- Difficult to support real-time applications
- Limited scalability for billions of IoT devices
- Cannot dynamically adapt to changing network conditions

# Future Networking Technologies

- **Software-Defined Networking (SDN):** Centralized and programmable network control.
- **Network Function Virtualization (NFV):** Network services run as software instead of dedicated hardware.
- **Artificial Intelligence (AI):** Predicts failures and optimizes traffic automatically.
- **5G & 6G Networks:** Ultra-low latency and high-speed communication.
- **Edge Computing:** Processes data closer to users for faster response.

## **Benefits:**

- Improved reliability
- Faster communication
- Better scalability
- Intelligent automation

# Future Reliable Networking Architecture

- **Key Features:**
- AI-based traffic management
- Self-healing networks
- Dynamic routing
- Zero Trust Security
- Cloud-native networking
- Intent-Based Networking (IBN)

# APPLICATION

- Smart Cities
- Autonomous Vehicles
- Industrial IoT
- Healthcare
- Smart Homes
- Online Gaming

# Conclusion & Future Scope

- **Conclusion**

- The traditional OSI model remains an important learning framework.
- Modern networks require intelligent, programmable, and adaptive architectures.
- Technologies like AI, SDN, NFV, Edge Computing, and 6G are shaping the future of reliable networking.

- **Future Scope**

- AI-driven autonomous networks
- Quantum networking
- Green and energy-efficient networking
- Secure Internet of Everything (IoE)



**THANKYOU!**

